06-Exercices-GARCH-VaR

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Exercice 1 : Modéliser le risque du CAC40

Données à partir de la base Yahoo Finance : fonction `pdfetch::pdfetch_YAHOO()`.

Vous avez pour mission de modéliser le risque présent sur le marché du CAC40.

Question 0

Importer le setup de l'analyse.

Question 1

Importer les données du CAC40 à partir de la base Yahoo Finance.

Question 2

Les données de cours boursier du CAC40 sont-elles stationnaires ?

Question 3

Stationnariser la série si besoin

Question 4

Modéliser le risque à partir d'une mesure de Value at Risk non-conditionnelle historique, de Student (normale pour une version plus simple) en statique et en fenêtre glissante de 252 jours. Commentez.

Question 5

Trouver le meilleur modèle (S)ARIMA pour modéliser l'espérance conditionnelle de la série stationnaire à partir du critère d'information BIC

Question 6

Les résidus de ce modèle sont-ils normaux ? Sont-ils auto-corrélés ? Les résidus au carré sont-ils auto-corrélés ? Y a-t-il des effets ARCH dans les résidus. Qu'en concluez-vous pour la suite de l'analyse.

Question 7

Estimer la volatilité conditionnelle à partir de 3 modèles (minimum) différents. Expliquez votre processus de sélection et de validation des modèles utilisés.

Question 8

Calculer la VaR associée à ces 3 modèles.

Question 9

Effectuer les tests de Backtests sur tous les modèles de VaR estimés. Quelles sont les conclusions ?

Question 10

Parmi tous les modèles de VaR validés (ou parmi tous les modèles s'il n'y a pas assez de modèles validés) quel est le modèle que vous choisissez pour modéliser le risque du CAC40 ?

Exercice 2 : Prédire le FTSE100

Votre patron vous demande de prédire les futurs rendements du FTSE100 ainsi que les risques de perte maximum pour un niveau de confiance de 99%. Expliquer votre démarche et abalyser les résultats.

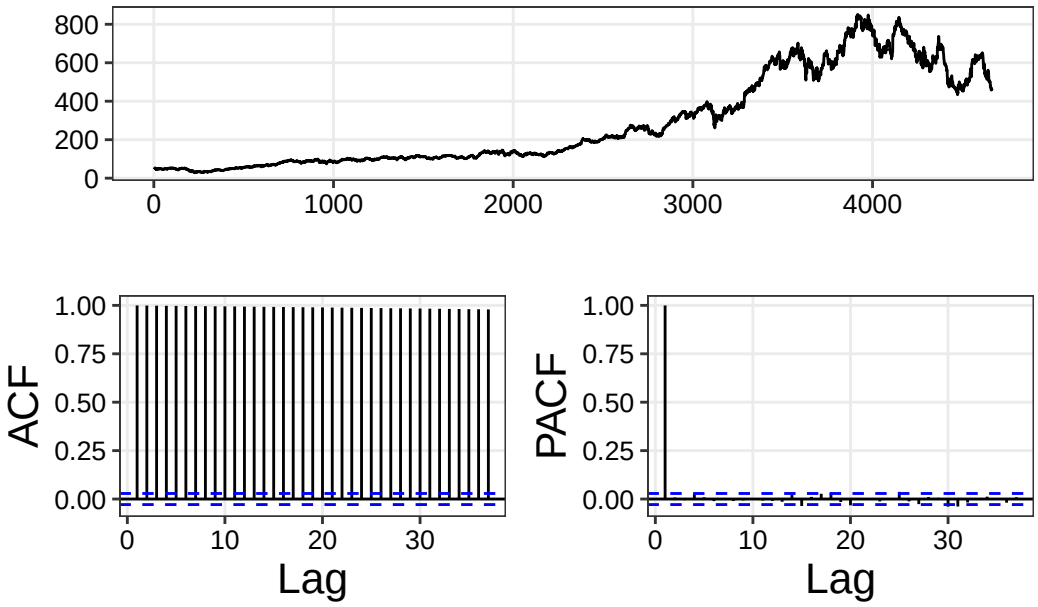
Exercice 3

Votre collège part à la retraite et vous transmet le projet sur lequel il travaillait : modéliser le risque des prix de l'action LVMH. Commentez les résultats et les méthodes qu'il utilise.

```
# Exécute le script setup.R pour charger tous les éléments importants
source(here::here("02-codes", "utils", "setup.R"))
```

```
lvmh <-
  pdfetch::pdfetch_YAHOO(
    "MC.PA", # Code de la série que l'on souhaite importer
    fields = "adjclose", # Variable que l'on souhaite importer
    from = as.Date("2008-01-03"), # Date de départ
    to = as.Date("2026-03-27") # Date de fin
  )
```

```
forecast::ggtsdisplay(lvmh, point = FALSE)
```




```
plot(lvmh)
```

```r
urca::ur.df(lvmh, "trend", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression trend


Call:
lm(formula = z.diff ~ z.lag.1 + 1 + tt + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max
-46.415  -1.581  -0.043   1.628  83.431

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)  -0.0069552  0.2142287  -0.032  0.97410
z.lag.1      -0.0016869  0.0009619  -1.754  0.07955 .
tt            0.0002516  0.0001712   1.470  0.14173
z.diff.lag1  -0.0154561  0.0146999  -1.051  0.29311
```

```
z.diff.lag2   0.0103342  0.0146878   0.704  0.48172
z.diff.lag3  -0.0393154  0.0146881  -2.677  0.00746 **
z.diff.lag4  -0.0074514  0.0146900  -0.507  0.61201
z.diff.lag5   0.0134215  0.0146851   0.914  0.36079
z.diff.lag6  -0.0062562  0.0146876  -0.426  0.67016
z.diff.lag7   0.0103928  0.0146877   0.708  0.47924
z.diff.lag8   0.0043106  0.0146892   0.293  0.76919
z.diff.lag9   0.0074864  0.0146896   0.510  0.61033
z.diff.lag10 -0.0079605  0.0147054  -0.541  0.58830
z.diff.lag11  0.0106389  0.0147066   0.723  0.46947
z.diff.lag12  0.0121690  0.0147075   0.827  0.40805
z.diff.lag13 -0.0250169  0.0147090  -1.701  0.08905 .
z.diff.lag14  0.0423764  0.0147134   2.880  0.00399 **
z.diff.lag15 -0.0126623  0.0147144  -0.861  0.38954
z.diff.lag16 -0.0415922  0.0147152  -2.826  0.00473 **
z.diff.lag17 -0.0400166  0.0147267  -2.717  0.00661 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.622 on 4624 degrees of freedom
Multiple R-squared:  0.00938,   Adjusted R-squared:  0.005309
F-statistic: 2.304 on 19 and 4624 DF,  p-value: 0.001053


Value of test-statistic is: -1.7537 1.388 1.5773

Critical values for test statistics:
      1pct  5pct 10pct
tau3 -3.96 -3.41 -3.12
phi2  6.09  4.68  4.03
phi3  8.27  6.25  5.34
```

```r
urca::ur.df(lvmh, "drift", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression drift


Call:
```

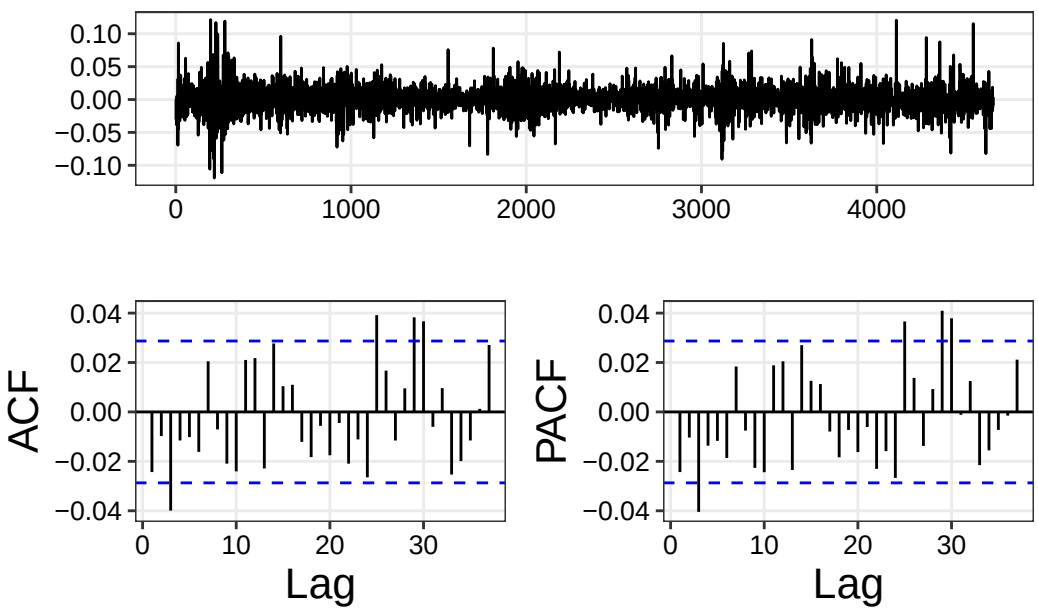
```
lm(formula = z.diff ~ z.lag.1 + 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-46.265  -1.577  -0.065   1.659  83.381 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)   0.2145522  0.1522562   1.409  0.15886    
z.lag.1      -0.0004062  0.0004073  -0.997  0.31869    
z.diff.lag1  -0.0162385  0.0146921  -1.105  0.26911    
z.diff.lag2   0.0095840  0.0146808   0.653  0.51390    
z.diff.lag3  -0.0400535  0.0146813  -2.728  0.00639 ** 
z.diff.lag4  -0.0082082  0.0146828  -0.559  0.57617    
z.diff.lag5   0.0126889  0.0146784   0.864  0.38738    
z.diff.lag6  -0.0069927  0.0146809  -0.476  0.63387    
z.diff.lag7   0.0096639  0.0146812   0.658  0.51041    
z.diff.lag8   0.0035920  0.0146829   0.245  0.80675    
z.diff.lag9   0.0067602  0.0146831   0.460  0.64525    
z.diff.lag10 -0.0086569  0.0146996  -0.589  0.55594    
z.diff.lag11  0.0099524  0.0147010   0.677  0.49845    
z.diff.lag12  0.0114888  0.0147021   0.781  0.43458    
z.diff.lag13 -0.0257159  0.0147032  -1.749  0.08036 .  
z.diff.lag14  0.0417033  0.0147081   2.835  0.00460 ** 
z.diff.lag15 -0.0133224  0.0147094  -0.906  0.36514    
z.diff.lag16 -0.0422469  0.0147103  -2.872  0.00410 ** 
z.diff.lag17 -0.0406475  0.0147223  -2.761  0.00579 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.623 on 4625 degrees of freedom
Multiple R-squared:  0.008917,  Adjusted R-squared:  0.00506 
F-statistic: 2.312 on 18 and 4625 DF,  p-value: 0.001299


Value of test-statistic is: -0.9973 1.0019 

Critical values for test statistics: 
      1pct  5pct 10pct
tau2 -3.43 -2.86 -2.57
phi1  6.43  4.59  3.78
```

```r
urca::ur.df(lvmh, "none", lags = 20, selectlags = "AIC") |> summary()
```

```
###############################################
# Augmented Dickey-Fuller Test Unit Root Test #
###############################################

Test regression none


Call:
lm(formula = z.diff ~ z.lag.1 - 1 + z.diff.lag)

Residuals:
    Min      1Q  Median      3Q     Max 
-46.310  -1.453   0.080   1.786  83.306 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)   
z.lag.1       0.0000351  0.0002605   0.135  0.89284   
z.diff.lag1  -0.0162521  0.0146936  -1.106  0.26876   
z.diff.lag2   0.0095760  0.0146824   0.652  0.51430   
z.diff.lag3  -0.0400645  0.0146829  -2.729  0.00638 **
z.diff.lag4  -0.0082023  0.0146844  -0.559  0.57648   
z.diff.lag5   0.0126976  0.0146800   0.865  0.38711   
z.diff.lag6  -0.0069850  0.0146824  -0.476  0.63428   
z.diff.lag7   0.0096731  0.0146827   0.659  0.51005   
z.diff.lag8   0.0035986  0.0146845   0.245  0.80642   
z.diff.lag9   0.0067622  0.0146847   0.460  0.64518   
z.diff.lag10 -0.0086566  0.0147012  -0.589  0.55600   
z.diff.lag11  0.0099564  0.0147026   0.677  0.49832   
z.diff.lag12  0.0114885  0.0147036   0.781  0.43464   
z.diff.lag13 -0.0257232  0.0147048  -1.749  0.08030 . 
z.diff.lag14  0.0417073  0.0147096   2.835  0.00460 **
z.diff.lag15 -0.0133383  0.0147110  -0.907  0.36462   
z.diff.lag16 -0.0422575  0.0147118  -2.872  0.00409 **
z.diff.lag17 -0.0406428  0.0147239  -2.760  0.00580 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.623 on 4626 degrees of freedom
Multiple R-squared:  0.008667,	Adjusted R-squared:  0.00481 
F-statistic: 2.247 on 18 and 4626 DF,  p-value: 0.001876
```

```
Value of test-statistic is: 0.1347

Critical values for test statistics:
      1pct  5pct 10pct
tau1 -2.58 -1.95 -1.62
```

```r
rend_lvmh <-
  log(lvmh) - log(lag(lvmh)) |>
  na.omit()

forecast::ggtsdisplay(rend_lvmh, point = FALSE)
```




```r
moments::jarque.test(as.numeric(rend_lvmh))
```

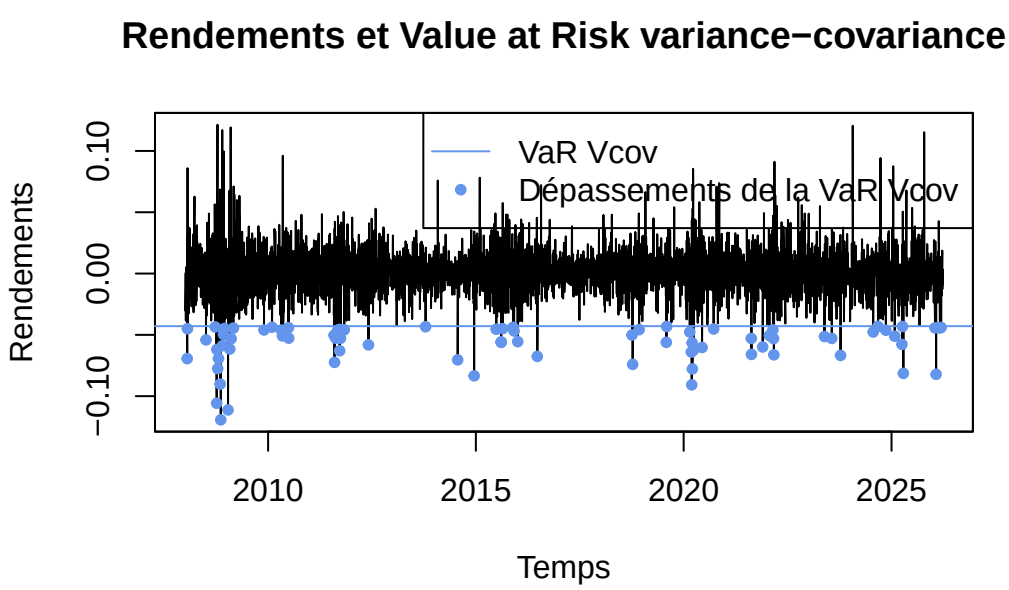
```
    Jarque-Bera Normality Test

data:  as.numeric(rend_lvmh)
JB = 4287, p-value < 0.00000000000000022
alternative hypothesis: greater
```

```r
t_couv <- 0.01
```

```r
VaR_VCov <- (-(mean(rend_lvmh)+sd(rend_lvmh)*qnorm(t_couv))) |> print()
```

```
[1] 0.04291614
```

```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk variance-covariance",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_VCov, col="cornflowerblue")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_VCov]), pch=20,
  col="cornflowerblue")
  legend("topright", legend = c("VaR Vcov","Dépassements de la VaR Vcov"),
         pch = c(NA,20), lty = c(1, NA), col = "cornflowerblue")
```

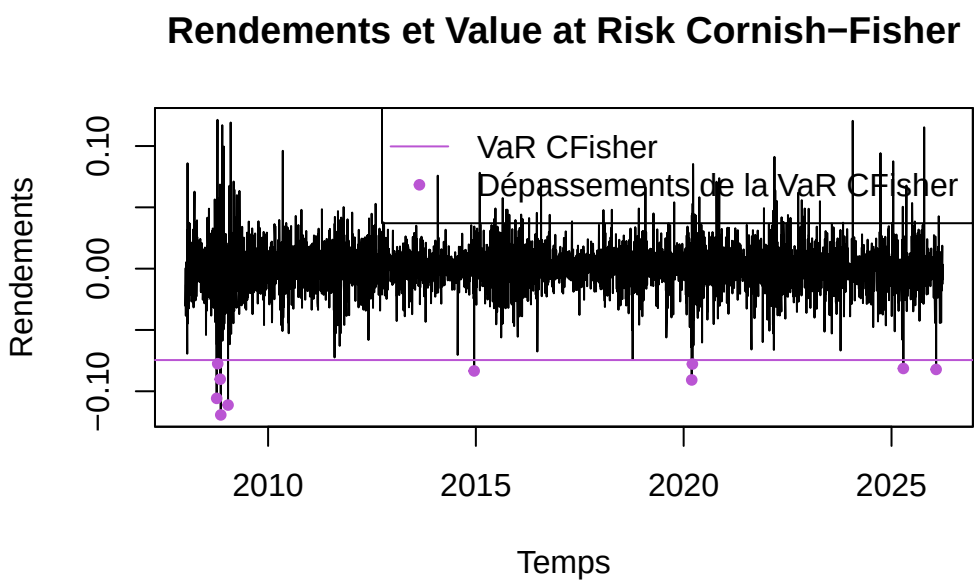



```r
K <- kurtosis(rend_lvmh)
S <- skewness(rend_lvmh)

k <-
as.numeric(qnorm(t_couv)+(qnorm(t_couv)^2-1)*(S/6)+(qnorm(t_couv)^3-3*qnorm(t_couv))*(K/24)+
```

```r
VaR_CFisher<- as.numeric(-(mean(rend_lvmh)+sd(rend_lvmh)*k)) |> print()
```

```
[1] 0.07455495
```

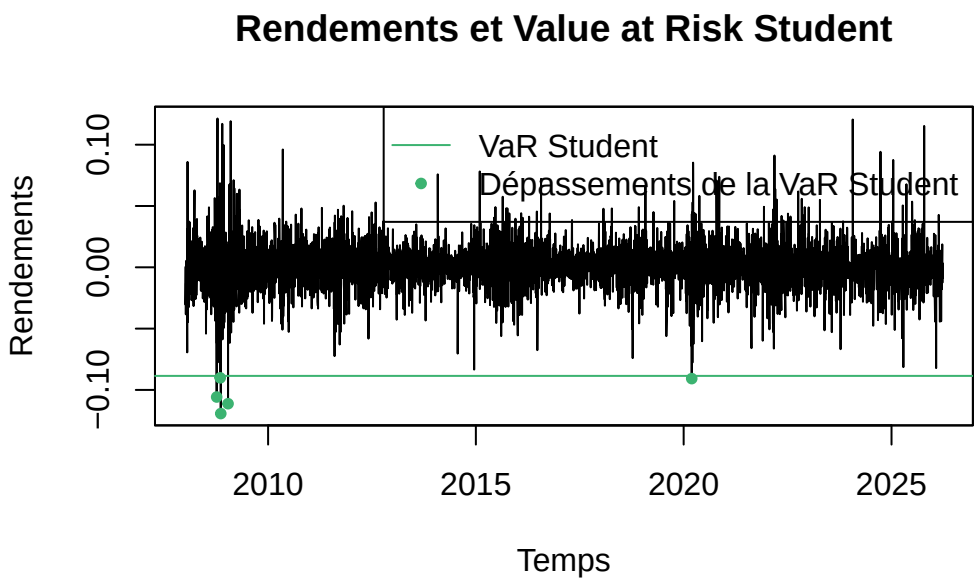
```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk Cornish-Fisher",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_CFisher, col="mediumorchid")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_CFisher]), pch=20,
  col="mediumorchid")
  legend("topright", legend = c("VaR CFisher","Dépassements de la VaR
  CFisher"),
         pch = c(NA,20), lty = c(1, NA), col = "mediumorchid")
```




```r
param.sd <- fitdist(distribution = 'std', x = rend_lvmh)$pars
quantile.sd <- qt(t_couv, df=param.sd["shape"])

VaR_student <-
(-mean(rend_lvmh)-sd(rend_lvmh)*quantile.sd*sqrt((quantile.sd-2)/quantile.sd))
|> print()
```

```
[1] 0.08855773
```

```r
plot(as.zoo(rend_lvmh),
      main = "Rendements et Value at Risk Student",
      xlab = "Temps",
      ylab = "Rendements")
  abline(h= -VaR_student, col="mediumseagreen")
  points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_student]), pch=20,
  col="mediumseagreen")
  legend("topright", legend = c("VaR Student","Dépassements de la VaR
  Student"),
         pch = c(NA,20), lty = c(1, NA), col = "mediumseagreen")
```




```r
n_backtest <- 2000

taille_sample_estimation <- length(rend_lvmh) - n_backtest

sample_estimation <- rend_lvmh[1:taille_sample_estimation]

sample_backtest <- rend_lvmh[(taille_sample_estimation+1):length(rend_lvmh)]
```

```r
VaR_VCov_estimation <-
(-(mean(sample_estimation)+sd(sample_estimation)*qnorm(t_couv)))

test_VaR_VCov <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_VCov_estimation, times = length(sample_backtest)),
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_VCov)),
  value = unlist(test_VaR_VCov)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   34
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       8.18194510909123
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00423093684953424
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       10.4255909694824
10 cc.critical     5.99146454710798
11 cc.LRp          0.00544642698307951
12 cc.Decision     Reject H0
```

```r
HIT_VCov <-
  if_else(sample_backtest < -VaR_VCov_estimation, 1, 0)

HIT_centre_VCov <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_VCov, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test
```

```
data:  HIT_centre_VCov
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_VCov,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.71827 -0.02552  0.00925  0.02990  0.73041

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762
L(HIT_centre, 2) -0.016735   0.022497  -0.744               0.4571
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264
L(HIT_centre, 4) -0.004360   0.022506  -0.194               0.8464
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185043   0.139890  -1.323               0.1861
L(rend, 2)       -0.317957   0.139848  -2.274               0.0231 *
L(rend, 3)        0.058906   0.140008   0.421               0.6740
L(rend, 4)        0.021289   0.140771   0.151               0.8798
L(rend, 5)       -0.024332   0.140435  -0.173               0.8625
```

```
rend2            76.422565   2.662759  28.701 < 0.0000000000000002 ***
L(rend2, 1)       0.211862   3.165890   0.067               0.9467
L(rend2, 2)       6.670556   3.144205   2.122               0.0340 *
L(rend2, 3)       0.909272   3.139157   0.290               0.7721
L(rend2, 4)      -2.119301   3.147436  -0.673               0.5008
L(rend2, 5)      -3.543882   3.145992  -1.126               0.2601
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,    Adjusted R-squared:  0.4192
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
K <- kurtosis(sample_estimation)
S <- skewness(sample_estimation)

k <-
as.numeric(qnorm(t_couv)+(qnorm(t_couv)^2-1)*(S/6)+(qnorm(t_couv)^3-3*qnorm(t_couv))*(K/24)+

VaR_CFisher_estimation <-
as.numeric(-(mean(sample_estimation)+sd(sample_estimation)*k))

test_VaR_CFisher <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_CFisher_estimation, times = length(sample_backtest)),
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_CFisher)),
  value = unlist(test_VaR_CFisher)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   4
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       19.2534427663781
```

```
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0000114463780930274
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       19.2694828773756
10 cc.critical     5.99146454710798
11 cc.LRp          0.0000654161466900893
12 cc.Decision     Reject H0
```

```r
HIT_CFisher <-
  if_else(sample_backtest < -VaR_CFisher_estimation, 1, 0)

HIT_centre_CFisher <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_VCov, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre_VCov
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_CFisher,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-0.71827 -0.02552  0.00925  0.02990  0.73041 

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)    
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762    
L(HIT_centre, 2) -0.016735   0.022497  -0.744               0.4571    
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264    
L(HIT_centre, 4) -0.004360   0.022506  -0.194               0.8464    
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485    
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185043   0.139890  -1.323               0.1861    
L(rend, 2)       -0.317957   0.139848  -2.274               0.0231 *  
L(rend, 3)        0.058906   0.140008   0.421               0.6740    
L(rend, 4)        0.021289   0.140771   0.151               0.8798    
L(rend, 5)       -0.024332   0.140435  -0.173               0.8625    
rend2            76.422565   2.662759  28.701 < 0.0000000000000002 ***
L(rend2, 1)       0.211862   3.165890   0.067               0.9467    
L(rend2, 2)       6.670556   3.144205   2.122               0.0340 *  
L(rend2, 3)       0.909272   3.139157   0.290               0.7721    
L(rend2, 4)      -2.119301   3.147436  -0.673               0.5008    
L(rend2, 5)      -3.543882   3.145992  -1.126               0.2601    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,	Adjusted R-squared:  0.4192 
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
param.sd <- fitdist(distribution = 'std', x = sample_estimation)$pars
quantile.sd <- qt(t_couv, df=param.sd["shape"])

VaR_student_estimation <-
(-mean(sample_estimation)-sd(sample_estimation)*quantile.sd*sqrt((quantile.sd-2)/quantile.sd

test_VaR_student <-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = rep(-VaR_student_estimation, times = length(sample_backtest)),
    conf.level = 0.95
```

```r
  )

tibble(
  name = names(unlist(test_VaR_student)),
  value = unlist(test_VaR_student)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   1
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       32.1902782785452
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00000001397884619169 79
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       32.1912792795877
10 cc.critical     5.99146454710798
11 cc.LRp          0.00000010227100455395 5
12 cc.Decision     Reject H0
```

```r
HIT_student <-
  if_else(sample_backtest < -VaR_student_estimation, 1, 0)

HIT_centre_student <- HIT_VCov - t_couv

stats::Box.test(HIT_centre_student, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre_student
X-squared = 28.608, df = 5, p-value = 0.00002767
```

```r
df <- tibble(
  HIT_centre = HIT_centre_student,
  rend = sample_backtest |> as.numeric(),
  rend2 = sample_backtest**2 |> as.numeric()
)
```

```r
ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

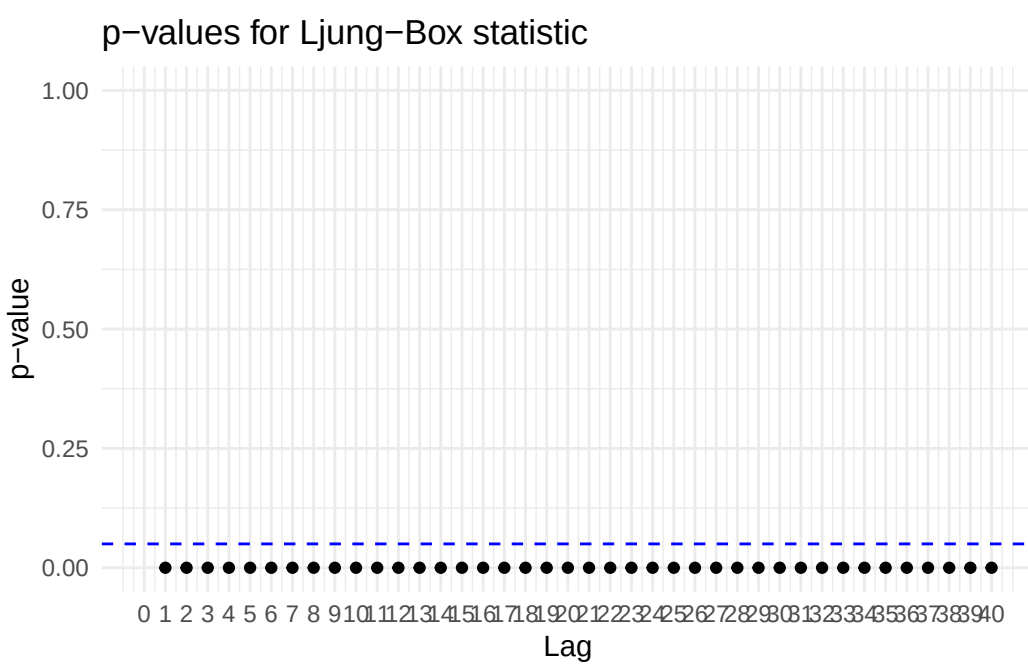
```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.71827 -0.02552  0.00925  0.02990  0.73041

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.018914   0.003056  -6.190       0.000000000731 ***
L(HIT_centre, 1)  0.003505   0.022494   0.156               0.8762
L(HIT_centre, 2) -0.016735   0.022497  -0.744               0.4571
L(HIT_centre, 3)  0.017929   0.022536   0.796               0.4264
L(HIT_centre, 4) -0.004360   0.022506  -0.194               0.8464
L(HIT_centre, 5)  0.004296   0.022482   0.191               0.8485
rend             -3.152929   0.120388 -26.190 < 0.0000000000000002 ***
L(rend, 1)       -0.185043   0.139890  -1.323               0.1861
L(rend, 2)       -0.317957   0.139848  -2.274               0.0231 *
L(rend, 3)        0.058906   0.140008   0.421               0.6740
L(rend, 4)        0.021289   0.140771   0.151               0.8798
L(rend, 5)       -0.024332   0.140435  -0.173               0.8625
rend2            76.422565   2.662759  28.701 < 0.0000000000000002 ***
L(rend2, 1)       0.211862   3.165890   0.067               0.9467
L(rend2, 2)       6.670556   3.144205   2.122               0.0340 *
L(rend2, 3)       0.909272   3.139157   0.290               0.7721
L(rend2, 4)      -2.119301   3.147436  -0.673               0.5008
L(rend2, 5)      -3.543882   3.145992  -1.126               0.2601
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.09867 on 1977 degrees of freedom
Multiple R-squared:  0.4241,    Adjusted R-squared:  0.4192
F-statistic: 85.64 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
best_arma <- forecast::auto.arima(rend_lvmh)

ar <- best_arma$arma[1]
ma <- ar <- best_arma$arma[2]

best_arma$arma[c(1,6,2)] |>  print()
```

```
[1] 1 0 2
```

```r
LSTS::Box.Ljung.Test(best_arma$residuals**2, lag = 40)
```




```r
fDMA::archtest(as.numeric(best_arma$residuals**2), lag = 20)
```

```
    Engle's LM ARCH Test

data:  as.numeric(best_arma$residuals^2)
statistic = 213.21, lag = 20, p-value < 0.00000000000000022
alternative hypothesis: ARCH effects of order 20 are present
```

```r
garch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "sGARCH",
        garchOrder = c(1,1),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

garch11_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_lvmh
  )

garch11_fit
```

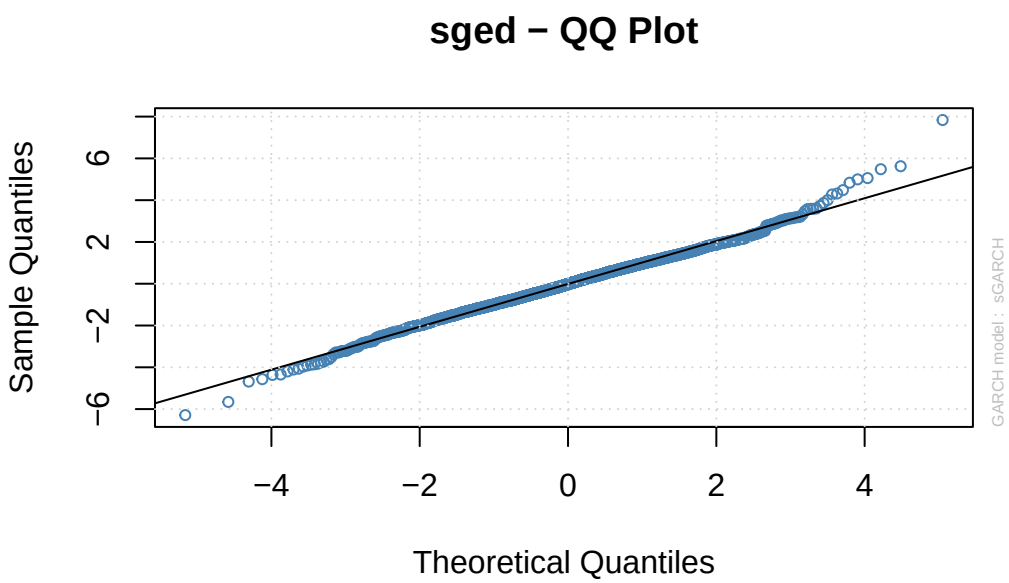
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : sGARCH(1,1)
Mean Model  : ARFIMA(2,0,2)
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error  t value Pr(>|t|)
mu      0.000703    0.000216   3.2524 0.001144
ar1     0.068234    0.017817   3.8297 0.000128
ar2     0.410813    0.193919   2.1185 0.034135
ma1    -0.106156    0.016397  -6.4742 0.000000
ma2    -0.429621    0.191680  -2.2413 0.025004
```

```
alpha1  0.052970    0.002687  19.7107 0.000000
beta1   0.930335    0.002450 379.7600 0.000000
skew    0.986794    0.019409  50.8427 0.000000
shape   1.227229    0.030140  40.7176 0.000000
omega   0.000006          NA       NA       NA

Robust Standard Errors:
        Estimate  Std. Error   t value Pr(>|t|)
mu      0.000703    0.000270   2.60040 0.009312
ar1     0.068234    0.094105   0.72509 0.468399
ar2     0.410813    0.482843   0.85082 0.394868
ma1    -0.106156    0.092742  -1.14464 0.252356
ma2    -0.429621    0.479414  -0.89614 0.370179
alpha1  0.052970    0.007485   7.07695 0.000000
beta1   0.930335    0.007011 132.70537 0.000000
skew    0.986794    0.024231  40.72384 0.000000
shape   1.227229    0.039168  31.33280 0.000000
omega   0.000006          NA        NA       NA

LogLikelihood : 12447.91

Information Criteria
------------------------------------

Akaike       -5.3340
Bayes        -5.3216
Shibata      -5.3340
Hannan-Quinn -5.3296

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.6109  0.4344
Lag[2*(p+q)+(p+q)-1][11]   5.3391  0.8655
Lag[4*(p+q)+(p+q)-1][19]   8.4402  0.7294
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                 0.00002628  0.9959
Lag[2*(p+q)+(p+q)-1][5] 0.35000649  0.9782
```

```
Lag[4*(p+q)+(p+q)-1][9] 1.18069516  0.9775
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]   0.09666 0.500 2.000  0.7559
ARCH Lag[5]   0.61766 1.440 1.667  0.8484
ARCH Lag[7]   0.93421 2.315 1.543  0.9240

Nyblom stability test
------------------------------------
Joint Statistic:  2.9792
Individual Statistics:
mu     0.2528
ar1    0.1295
ar2    0.1796
ma1    0.1471
ma2    0.1956
alpha1 0.4905
beta1  0.6455
skew   0.1343
shape  1.4007

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.1 2.32 2.82
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value     prob sig
Sign Bias            1.579 0.114492
Negative Sign Bias   1.599 0.109891
Positive Sign Bias   2.699 0.006981 ***
Joint Effect        10.505 0.014726  **


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     33.19      0.02287
2    30     33.08      0.27436
3    40     47.75      0.15883
```

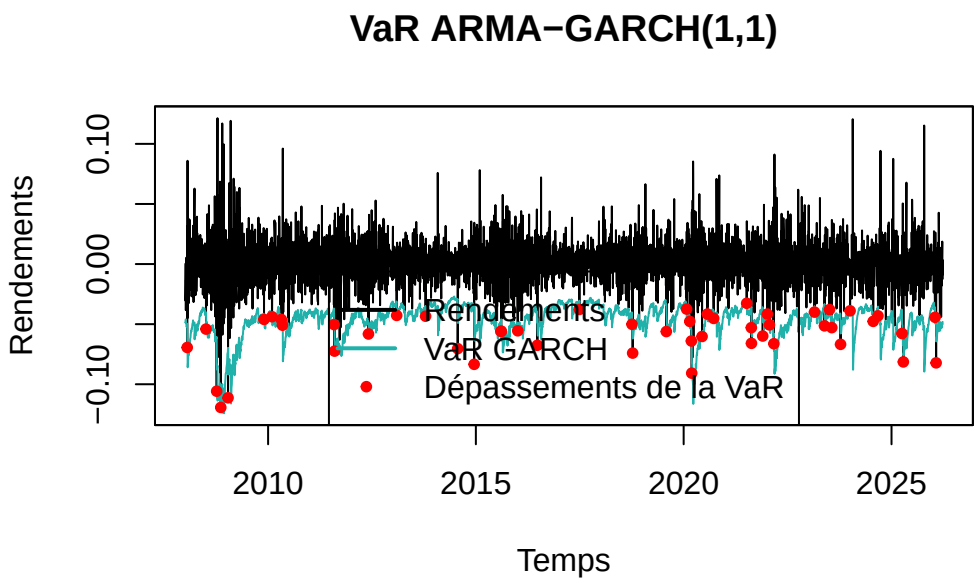
```
4     50      60.59       0.12388

Elapsed time : 1.041143
```

```r
plot(garch11_fit, which = 9)
```




```r
# Calculer la VaR à partir du modèle EGARCH
VaR_GARCH <- -(as.numeric(quantile(garch11_fit, probs = t_couv)))

# Représenter les rendements et la VaR
plot(
  # Combiner dans un seul df les rendements et la VaR
  as.zoo(cbind(rend_lvmh,-VaR_GARCH)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-GARCH(1,1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_GARCH]), pch=20,
col="red")
```

```
legend("bottom",c("Rendements","VaR GARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
garch11_estimation_fit <-
  rugarch::ugarchfit(
    garch11_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_garch11 <-
  rugarch::ugarchforecast(
    garch11_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_GARCH_backtest <- -(as.numeric(quantile(forecast_garch11, probs = 0.01)))

test_VaR_GARCH<-
```

```r
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_GARCH_backtest,
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_GARCH)),
  value = unlist(test_VaR_GARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   30
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       4.37849677769573
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.0363950816720029
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       7.41028267304434
10 cc.critical     5.99146454710798
11 cc.LRp          0.0245967407041665
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_GARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 7.8284, df = 5, p-value = 0.166
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.58429 -0.02453  0.00551  0.02254  0.84485

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.008767   0.002990  -2.932              0.00341 **
L(HIT_centre, 1)  0.009453   0.022519   0.420              0.67469
L(HIT_centre, 2) -0.028049   0.022499  -1.247              0.21267
L(HIT_centre, 3)  0.004019   0.022500   0.179              0.85825
L(HIT_centre, 4) -0.003649   0.022482  -0.162              0.87109
L(HIT_centre, 5)  0.006731   0.022462   0.300              0.76445
rend             -2.644756   0.123224 -21.463 < 0.0000000000000002 ***
L(rend, 1)       -0.187270   0.137030  -1.367              0.17190
L(rend, 2)       -0.202942   0.136924  -1.482              0.13846
L(rend, 3)       -0.176487   0.136651  -1.292              0.19668
L(rend, 4)        0.011666   0.137550   0.085              0.93242
L(rend, 5)        0.063204   0.137068   0.461              0.64477
rend2            62.710722   2.715540  23.093 < 0.0000000000000002 ***
L(rend2, 1)       0.657877   3.056802   0.215              0.82962
L(rend2, 2)      -5.265746   3.034859  -1.735              0.08288 .
```

```
L(rend2, 3)      -4.618026   3.029828  -1.524              0.12762
L(rend2, 4)      -5.419404   3.037486  -1.784              0.07455 .
L(rend2, 5)      -5.051536   3.041042  -1.661              0.09685 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1007 on 1977 degrees of freedom
Multiple R-squared:  0.3218,    Adjusted R-squared:  0.3159
F-statistic: 55.17 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
egarch11_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "eGARCH",
        garchOrder = c(1,1),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

egarch11_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_lvmh
  )

egarch11_fit
```

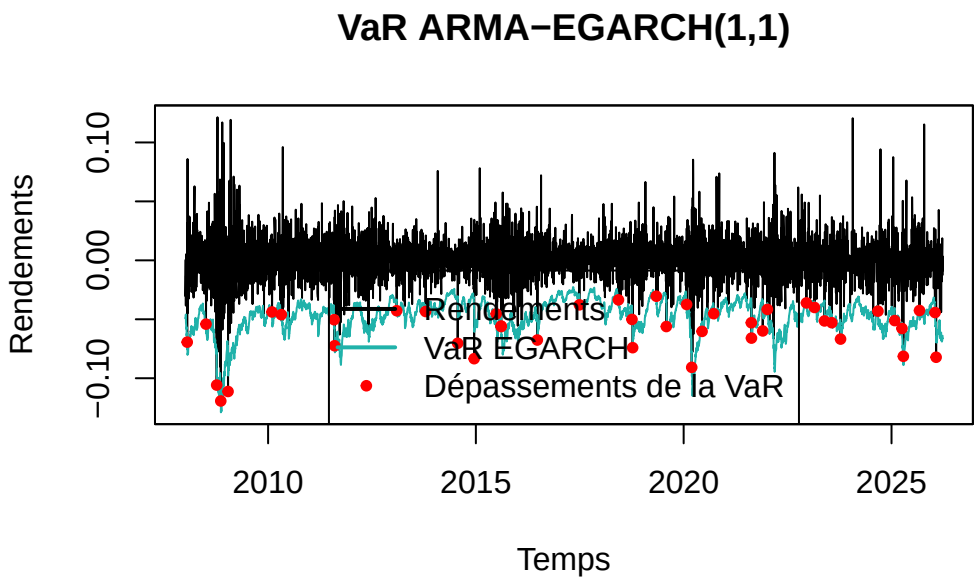
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : eGARCH(1,1)
Mean Model  : ARFIMA(2,0,2)
```

```
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error    t value Pr(>|t|)
mu      0.000168    0.000161     1.0435   0.2967
ar1     1.157565    0.039544    29.2728   0.0000
ar2    -0.638229    0.016707   -38.2022   0.0000
ma1    -1.174158    0.041426   -28.3434   0.0000
ma2     0.628004    0.015800    39.7482   0.0000
alpha1 -0.082090    0.006083   -13.4940   0.0000
beta1   0.983923    0.000025 39250.4705   0.0000
gamma1  0.083801    0.007746    10.8193   0.0000
skew    0.972285    0.016562    58.7056   0.0000
shape   1.277397    0.034566    36.9555   0.0000
omega  -0.128071          NA         NA       NA

Robust Standard Errors:
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000168    0.000175     0.96513  0.33448
ar1     1.157565    0.012750    90.78673  0.00000
ar2    -0.638229    0.005375  -118.74088  0.00000
ma1    -1.174158    0.013548   -86.66703  0.00000
ma2     0.628004    0.005811   108.07258  0.00000
alpha1 -0.082090    0.007397   -11.09711  0.00000
beta1   0.983923    0.000025 39356.75747  0.00000
gamma1  0.083801    0.009284     9.02606  0.00000
skew    0.972285    0.018028    53.93343  0.00000
shape   1.277397    0.042680    29.92981  0.00000
omega  -0.128071          NA          NA       NA

LogLikelihood : 12506.33

Information Criteria
------------------------------------

Akaike       -5.3586
Bayes        -5.3448
Shibata      -5.3586
Hannan-Quinn -5.3538

Weighted Ljung-Box Test on Standardized Residuals
------------------------------------
```

```
                          statistic p-value
Lag[1]                      0.01455   0.904
Lag[2*(p+q)+(p+q)-1][11]   2.80906   1.000
Lag[4*(p+q)+(p+q)-1][19]   5.94159   0.973
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                        statistic p-value
Lag[1]                     0.3301  0.5656
Lag[2*(p+q)+(p+q)-1][5]    0.9621  0.8682
Lag[4*(p+q)+(p+q)-1][9]    2.0757  0.8958
d.o.f=2

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[3]    0.1746 0.500 2.000  0.6760
ARCH Lag[5]    0.9098 1.440 1.667  0.7599
ARCH Lag[7]    1.3460 2.315 1.543  0.8517

Nyblom stability test
------------------------------------
Joint Statistic:  2.9163
Individual Statistics:
mu     0.1622
ar1    0.2226
ar2    0.1268
ma1    0.2260
ma2    0.1172
alpha1 0.4445
beta1  0.4628
gamma1 0.1541
skew   0.1420
shape  1.2126

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.29 2.54 3.05
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
```

```
                   t-value    prob sig
Sign Bias           2.1587 0.03093  **
Negative Sign Bias  0.8753 0.38143
Positive Sign Bias  2.0693 0.03858  **
Joint Effect        6.0220 0.11054


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     29.55       0.0578
2    30     37.50       0.1339
3    40     45.86       0.2089
4    50     57.46       0.1904


Elapsed time : 1.807327
```

```r
VaR_EGARCH <- -(as.numeric(quantile(egarch11_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_lvmh,-VaR_EGARCH)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-EGARCH(1,1)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_EGARCH]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR EGARCH", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
egarch11_estimation_fit <-
  rugarch::ugarchfit(
    egarch11_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_egarch11 <-
  rugarch::ugarchforecast(
    egarch11_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_EGARCH_backtest <- -(as.numeric(quantile(forecast_egarch11, probs =
0.01)))

test_VaR_EGARCH<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_EGARCH_backtest,
```

```r
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_EGARCH)),
  value = unlist(test_VaR_EGARCH)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   33
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       7.13670995390553
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00755215918452179
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       7.48366413009826
10 cc.critical     5.99146454710798
11 cc.LRp          0.0237106239097272
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_EGARCH_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 2.3334, df = 5, p-value = 0.8013
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
```

```r
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.60796 -0.02566  0.00439  0.02278  0.85526

Coefficients:
                  Estimate Std. Error t value            Pr(>|t|)
(Intercept)      -0.006437   0.003062  -2.102              0.0357 *
L(HIT_centre, 1) -0.006197   0.022501  -0.275              0.7830
L(HIT_centre, 2) -0.023667   0.022487  -1.052              0.2927
L(HIT_centre, 3)  0.021837   0.022457   0.972              0.3310
L(HIT_centre, 4)  0.001126   0.022444   0.050              0.9600
L(HIT_centre, 5)  0.002927   0.022082   0.133              0.8946
rend             -2.744581   0.127168 -21.582 <0.0000000000000002 ***
L(rend, 1)       -0.006976   0.141556  -0.049              0.9607
L(rend, 2)       -0.147915   0.141509  -1.045              0.2960
L(rend, 3)        0.006972   0.141487   0.049              0.9607
L(rend, 4)        0.050008   0.142374   0.351              0.7254
L(rend, 5)        0.027815   0.142132   0.196              0.8449
rend2            64.778815   2.808041  23.069 <0.0000000000000002 ***
L(rend2, 1)      -1.501092   3.159775  -0.475              0.6348
L(rend2, 2)      -5.763415   3.135990  -1.838              0.0662 .
L(rend2, 3)      -8.022125   3.132234  -2.561              0.0105 *
L(rend2, 4)      -4.552000   3.144439  -1.448              0.1479
L(rend2, 5)      -6.167839   3.140955  -1.964              0.0497 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1041 on 1977 degrees of freedom
Multiple R-squared:  0.3195,    Adjusted R-squared:  0.3136
F-statistic:  54.6 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
egarch22_spec <-
  rugarch::ugarchspec(
    variance.model =
      list(
        model = "eGARCH",
        garchOrder = c(2,2),
        variance.targeting = TRUE
      ),
    mean.model =
      list(
        armaOrder = c(ar,ma),
        include.mean = TRUE
      ),
    distribution.model = "sged"
  )

egarch22_fit <-
  rugarch::ugarchfit(
    egarch22_spec,
    rend_lvmh
  )

egarch22_fit
```

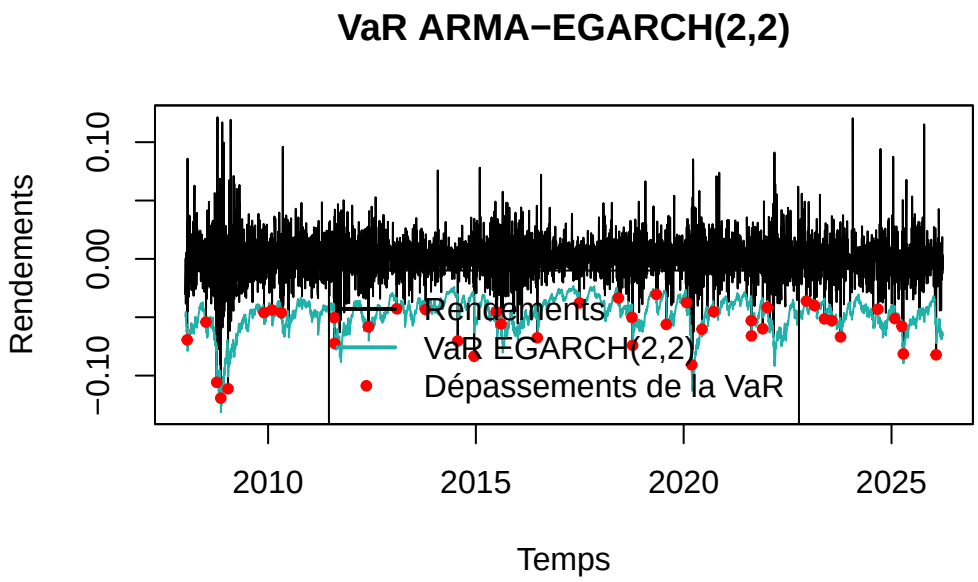
```
*---------------------------------*
*          GARCH Model Fit        *
*---------------------------------*

Conditional Variance Dynamics
-----------------------------------
GARCH Model : eGARCH(2,2)
Mean Model  : ARFIMA(2,0,2)
Distribution    : sged

Optimal Parameters
------------------------------------
        Estimate  Std. Error     t value Pr(>|t|)
```

```
mu      0.000194    0.000166     1.16937 0.242255
ar1    -0.358645    0.048324    -7.42162 0.000000
ar2     0.578222    0.027140    21.30549 0.000000
ma1     0.328444    0.047306     6.94295 0.000000
ma2    -0.605543    0.027415   -22.08838 0.000000
alpha1 -0.097091    0.025445    -3.81580 0.000136
alpha2  0.019435    0.026844     0.72401 0.469060
beta1   0.999995    0.000021 46954.87721 0.000000
beta2  -0.015082    0.002054    -7.34430 0.000000
gamma1  0.094821    0.036395     2.60529 0.009180
gamma2 -0.014420    0.033921    -0.42510 0.670762
skew    0.973378    0.018509    52.59065 0.000000
shape   1.279433    0.034453    37.13533 0.000000
omega  -0.120193          NA          NA       NA

Robust Standard Errors:
        Estimate  Std. Error     t value Pr(>|t|)
mu      0.000194    0.000169     1.14404 0.252608
ar1    -0.358645    0.011149   -32.16781 0.000000
ar2     0.578222    0.006184    93.50595 0.000000
ma1     0.328444    0.012240    26.83411 0.000000
ma2    -0.605543    0.006308   -96.00173 0.000000
alpha1 -0.097091    0.027562    -3.52270 0.000427
alpha2  0.019435    0.030020     0.64740 0.517370
beta1   0.999995    0.000024 41702.48520 0.000000
beta2  -0.015082    0.002401    -6.28084 0.000000
gamma1  0.094821    0.036935     2.56723 0.010251
gamma2 -0.014420    0.034640    -0.41628 0.677209
skew    0.973378    0.022377    43.49851 0.000000
shape   1.279433    0.042647    30.00079 0.000000
omega  -0.120193          NA          NA       NA

LogLikelihood : 12504.79

Information Criteria
------------------------------------

Akaike       -5.3567
Bayes        -5.3387
Shibata      -5.3567
Hannan-Quinn -5.3504

Weighted Ljung-Box Test on Standardized Residuals
```

```
------------------------------------
                         statistic p-value
Lag[1]                      0.5839  0.4448
Lag[2*(p+q)+(p+q)-1][11]    3.4448  1.0000
Lag[4*(p+q)+(p+q)-1][19]    6.5340  0.9435
d.o.f=4
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals
------------------------------------
                         statistic p-value
Lag[1]                       0.120  0.7290
Lag[2*(p+q)+(p+q)-1][11]     3.116  0.8547
Lag[4*(p+q)+(p+q)-1][19]     7.545  0.7260
d.o.f=4

Weighted ARCH LM Tests
------------------------------------
            Statistic Shape Scale P-Value
ARCH Lag[5]     0.914 0.500 2.000  0.3390
ARCH Lag[7]     1.470 1.473 1.746  0.6304
ARCH Lag[9]     2.794 2.402 1.619  0.6040

Nyblom stability test
------------------------------------
Joint Statistic:  4.6064
Individual Statistics:
mu     0.16929
ar1    0.06789
ar2    0.01572
ma1    0.07129
ma2    0.01769
alpha1 0.42332
alpha2 0.43024
beta1  0.43378
beta2  0.41725
gamma1 0.17353
gamma2 0.45957
skew   0.15217
shape  1.15067

Asymptotic Critical Values (10% 5% 1%)
Joint Statistic:         2.89 3.15 3.69
```

```
Individual Statistic:    0.35 0.47 0.75

Sign Bias Test
------------------------------------
                   t-value    prob sig
Sign Bias           1.6403 0.10102
Negative Sign Bias  0.2114 0.83259
Positive Sign Bias  1.7498 0.08022   *
Joint Effect        4.0313 0.25810


Adjusted Pearson Goodness-of-Fit Test:
------------------------------------
  group statistic p-value(g-1)
1    20     26.79       0.1097
2    30     45.97       0.0236
3    40     43.44       0.2877
4    50     50.28       0.4225


Elapsed time : 2.928868
```

```r
VaR_EGARCH22 <- -(as.numeric(quantile(egarch22_fit, probs = t_couv)))

plot(
  as.zoo(cbind(rend_lvmh,-VaR_EGARCH22)),
  screens=1,
  col = c("black","lightseagreen"),
  main="VaR ARMA-EGARCH(2,2)",
  xlab = "Temps",
  ylab = "Rendements"
)
points(as.zoo(rend_lvmh[as.numeric(rend_lvmh) < -VaR_EGARCH22]), pch=20,
col="red")
legend("bottom",c("Rendements","VaR EGARCH(2,2)", "Dépassements de la VaR"),
         col = c("black","lightseagreen", 'red'),lty = c(1,1,NA),lwd =
         c(2,2,NA),pch = c(NA,NA,20))
```




```
egarch22_estimation_fit <-
  rugarch::ugarchfit(
    egarch22_spec,
    rend_lvmh,
    out.sample = n_backtest
  )

forecast_egarch22 <-
  rugarch::ugarchforecast(
    egarch22_estimation_fit,
    data = rend_lvmh,
    n.roll = n_backtest-1,
    n.ahead = 1
  )

VaR_EGARCH22_backtest <- -(as.numeric(quantile(forecast_egarch22, probs =
0.01)))

test_VaR_EGARCH22<-
  rugarch::VaRTest(
    alpha = t_couv,
    actual = sample_backtest,
    VaR = -VaR_EGARCH22_backtest,
```

```r
    conf.level = 0.95
  )

tibble(
  name = names(unlist(test_VaR_EGARCH22)),
  value = unlist(test_VaR_EGARCH22)
) |>
  print(n = 20)
```

```
# A tibble: 12 x 2
   name            value
   <chr>           <chr>
 1 expected.exceed 20
 2 actual.exceed   33
 3 uc.H0           Correct Exceedances
 4 uc.LRstat       7.13670995390553
 5 uc.critical     3.84145882069412
 6 uc.LRp          0.00755215918452179
 7 uc.Decision     Reject H0
 8 cc.H0           Correct Exceedances & Independent
 9 cc.LRstat       13.0095513372679
10 cc.critical     5.99146454710798
11 cc.LRp          0.00149627638279803
12 cc.Decision     Reject H0
```

```r
HIT <- if_else(sample_backtest < -VaR_EGARCH22_backtest, 1, 0)

HIT_centre <- HIT - 0.01

stats::Box.test(HIT_centre, lag = 5, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  HIT_centre
X-squared = 13.528, df = 5, p-value = 0.01891
```

```r
df <-
  tibble(
    HIT_centre = HIT_centre,
    rend = sample_backtest,
    rend2 = sample_backtest**2
```

```r
  )

ARDL::ardl(
  HIT_centre ~ rend + rend2,
  data = df,
  order = 5
) |>
  summary()
```

```
Time series regression with "ts" data:
Start = 6, End = 2000

Call:
dynlm::dynlm(formula = full_formula, data = data, start = start,
    end = end)

Residuals:
     Min       1Q   Median       3Q      Max
-0.59603 -0.02628  0.00448  0.02283  0.85761

Coefficients:
                  Estimate Std. Error t value             Pr(>|t|)
(Intercept)      -0.006077   0.003072  -1.978              0.04807 *
L(HIT_centre, 1)  0.067740   0.022488   3.012              0.00263 **
L(HIT_centre, 2) -0.029915   0.022525  -1.328              0.18430
L(HIT_centre, 3)  0.021857   0.022492   0.972              0.33129
L(HIT_centre, 4) -0.002501   0.022498  -0.111              0.91149
L(HIT_centre, 5) -0.007926   0.022080  -0.359              0.71965
rend             -2.707469   0.127641 -21.212 < 0.0000000000000002 ***
L(rend, 1)        0.049106   0.141556   0.347              0.72870
L(rend, 2)       -0.103390   0.141398  -0.731              0.46475
L(rend, 3)       -0.059408   0.141305  -0.420              0.67422
L(rend, 4)        0.107224   0.142203   0.754              0.45092
L(rend, 5)       -0.048212   0.142094  -0.339              0.73442
rend2            63.585414   2.817451  22.568 < 0.0000000000000002 ***
L(rend2, 1)      -3.362486   3.154569  -1.066              0.28659
L(rend2, 2)      -5.519608   3.129887  -1.764              0.07797 .
L(rend2, 3)      -8.235363   3.125707  -2.635              0.00849 **
L(rend2, 4)      -4.508695   3.138588  -1.437              0.15101
L(rend2, 5)      -5.205372   3.136198  -1.660              0.09712 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.1045 on 1977 degrees of freedom
Multiple R-squared:  0.3147,    Adjusted R-squared:  0.3088
F-statistic:  53.4 on 17 and 1977 DF,  p-value: < 0.00000000000000022
```

```r
df_VaR <-
  tibble(
    VaR_VCov = VaR_VCov,
    VaR_CFisher = VaR_CFisher,
    VaR_student = VaR_student,
    VaR_GARCH= VaR_GARCH,
    VaR_EGARCH = VaR_EGARCH,
    VaR_EGARCH22 = VaR_EGARCH22
  ) |>
  drop_na()

df_VaR_long <-
  df_VaR |>
  mutate(temps = row_number()) |>
  pivot_longer(
    cols = -temps,
    names_to = "model",
    values_to = "VaR"
  ) |>
  mutate(
    .by = temps,
    moyenne_VaR = mean(VaR)
  ) |>
  mutate(
    ecart = (VaR - moyenne_VaR) / moyenne_VaR
  ) |>
  summarize(
    .by = model,
    MRB = abs(mean(ecart))
  ) |>
  print()
```

```
# A tibble: 6 x 2
  model          MRB
  <chr>        <dbl>
1 VaR_VCov     0.251
2 VaR_CFisher  0.301
3 VaR_student  0.546
```

```
4 VaR_GARCH    0.204
5 VaR_EGARCH   0.196
6 VaR_EGARCH22 0.197
```